

Kubespray Kubernetes

1. Kubespray

Kubespray Ansible Kubernetes Kubernetes Kubespray
 Docker containerd CRI-O Calico Flannel Cilium

2.

2.1

- 2GB 2 CPU
- 1GB 1 CPU
- 20GB

2.2

- Ansible 2.15
- Python 3.9
- Ubuntu 20.04/22.04 CentOS 7/8 Debian 10/11

2.3

-
- 6443
-

3.

3.1

```
submodules/kubespray/roles/kubespray-defaults/defaults/main/download.yml
```

```
# gcr and kubernetes image repo define
```

```
gcr_image_repo: "gcr.m.daocloud.io"
```

```
kube_image_repo: "k8s.m.daocloud.io"
```

```
# docker image repo define
```

```
docker_image_repo: "docker.m.daocloud.io"
```

```
# quay image repo define
```

```
quay_image_repo: "quay.m.daocloud.io"
```

```
# github image repo define (ex multus only use that)
```

```
github_image_repo: "ghcr.m.daocloud.io"
```

```
# Download URLs
dl_k8s_io_url: "https://kubernetes-release.pek3b.qingstor.com"
```

3.2 Docker

```
submodules/kubespray/roles/container-engine/docker/defaults/main.yml
```

```
# Docker registry mirrors
docker_registry_mirrors:
  - "https://docker.m.daocloud.io"
  - "https://registry.docker-cn.com"
  - "https://hub-mirror.c.163.com"
  - "https://docker.aityp.com/"
```

3.3 Kubernetes

```
submodules/kubespray/inventory/sample/group_vars/k8s_cluster/k8s-cluster.yml
```

```
# Kubernetes
kube_apiserver_image_repo: "{{ kube_image_repo }}/kube-apiserver"
kube_controller_manager_image_repo: "{{ kube_image_repo }}/kube-controller-manager"
kube_scheduler_image_repo: "{{ kube_image_repo }}/kube-scheduler"
kube_proxy_image_repo: "{{ kube_image_repo }}/kube-proxy"
etcd_image_repo: "{{ kube_image_repo }}/etcd"
coredns_image_repo: "{{ kube_image_repo }}/coredns/coredns"
pause_image_repo: "{{ kube_image_repo }}/pause"

# pod_infra_image_repo
pod_infra_image_repo: "{{ pause_image_repo }}"

# QingStor Kubernetes
kubelet_download_url: "https://kubernetes-release.pek3b.qingstor.com/release/{{ kube_version }}"
kubeadm_download_url: "https://kubernetes-release.pek3b.qingstor.com/release/{{ kube_version }}"
kubectldownload_url: "https://kubernetes-release.pek3b.qingstor.com/release/{{ kube_version }}"
```

4. Kubernetes

4.1 inventory

```
inventory
cp -r submodules/kubespray/inventory/sample inventory/mycluster
inventory/mycluster/hosts.yaml
```

```

all:
  hosts:
    master1:
      ansible_host: 192.168.0.200
      ip: 192.168.0.200
      access_ip: 192.168.0.200
    node1:
      ansible_host: 192.168.0.201
      ip: 192.168.0.201
      access_ip: 192.168.0.201
  children:
    kube_control_plane:
      hosts:
        master1:
    kube_node:
      hosts:
        master1:
        node1:
    etcd:
      hosts:
        master1:
    k8s_cluster:
      children:
        kube_control_plane:
        kube_node:
    calico_rr:
      hosts:

```

4.2

Ansible

```
ansible-playbook -i inventory/mycluster/hosts.yaml submodules/kubespray/cluster.yml -b
```

4.3

Ansible

5.

5.1

```

1.          3 2.          3.          bash    sudo docker pull
k8s.m.daocloud.io/pause:3.10

```

5.2 RBAC

kubernetes-admin

```
1. kubernetes-admin bash sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf
get clusterrolebindings | grep admin 2. cluster-admin bash
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf create
clusterrolebinding kubernetes-admin-cluster-admin --clusterrole=cluster-admin
--user=kubernetes-admin
```

5.3 Pause

kubelet registry.k8s.io/pause:3.10

```
1. kubeadm-setup.yml pause yaml - name: Kubeadm |
Pull and tag pause image command: >- /usr/local/bin/ctr
image pull {{ pod_infra_image_repo }}:{{ pod_infra_image_tag }}
&& /usr/local/bin/ctr image tag {{ pod_infra_image_repo
}}:{{ pod_infra_image_tag }} registry.k8s.io/pause:{{ pod_infra_image_tag
}}
become: true when: - inventory_hostname ==
first_kube_control_plane - not kubeadm_already_run.stat.exists
2. pod_infra_image_repo yaml pod_infra_image_repo: "{{{
pause_image_repo }}" pause_image_repo: "{{ kube_image_repo
}}/pause"
```

5.4 Containerd

containerd sandbox_image

```
1. containerd bash sudo containerd config default >
/etc/containerd/config.toml 2. sandbox toml [plugins."io.containerd.cri.v1.images"]
sandbox = "k8s.m.daocloud.io/pause:3.10" 3. containerd bash
sudo systemctl restart containerd
```

5.5 Kubelet

kubelet registry.k8s.io/pause:3.10

```
1. kubelet bash sudo vi /etc/kubernetes/kubelet.env 2.
--pod-infra-container-image bash KUBELET_ARGS="--bootstrap-kubeconfig=/etc/kubernetes/b
\ --pod-infra-container-image=k8s.m.daocloud.io/pause:3.10 \
--config=/etc/kubernetes/kubelet-config.yaml \ --kubeconfig=/etc/kubernetes/kubelet.conf
\ --runtime-cgroups=/system.slice/containerd.service \ " 3.
kubelet bash sudo systemctl restart kubelet
```

6.

6.1 API

```
sudo netstat -tulpn | grep 6443
```

API 6443

6.2

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf get nodes  
Ready
```

6.3

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf get pods -n kube-system  
kube-apiserver kube-controller-manager kube-scheduler etcd Running
```

6.4

```
sudo kubectl --kubeconfig=/etc/kubernetes/admin.conf version  
Kubernetes
```

7.

Kubespary	Kubernetes	Kubernetes
kubectl	Kubernetes	Pod Deployment Service